



1. If the direction North-East becomes South-East, how will other directions change ?

(Direction Distance) NIMCET-2010

- (a) West to North (b) South to South-West
(c) North-West to East (d) East to South-West
2. A man runs 20 m towards east and turns right, runs 10 m and turns right, runs 9 m and turns left, runs 5 m and turns left, runs 12 m and finally turns left and runs, 6 m. Which direction is the man facing? **(Direction and Distance) NIMCET-2012**

- (a) North
(b) South
(c) East
(d) West

3. A, B, C, D, E, F and G are sitting in a line facing East. C is immediate to the right of D. B is at one of the extreme ends has E as his neighbour. G is between E and F. D is sitting third from the South end. Who is sitting third from North? **(Direction and Distance) NIMCET-2012**

- (a) A
(b) E
(c) F
(d) G

4. Dhoni starts from his office at 8 am on a Sunday morning, travels 10 km towards West and then turns to his left and walks 8 km. Then, he again turns to his left and walks 4 km and then stops. What is the shortest distance to his office from the point where he stopped? **(Direction and Distance) NIMCET 2013**

- (a) 18 km (b) 8 km (c) 10 km (d) None

5. A road network has parallel roads, which are equidistant from each other and running North-South or East-West only. The road junctions A, B, C, H and X are such that A is East of B and West of C, H is South-west of C and South-East of B. B is South-East of X. Which of the junction are the farthest South. And the farthest East?

(Direction and Distance) NIMCET-2014

- (a) H, B (b) H, C
(c) C, H (d) B, H

Direction [Q. Nos. 6 to 10] : All the roads of a city are either perpendicular or parallel to one another. The roads are all straight. Roads A, B, C, D and E are parallel to one another. Roads F, G, H, I, J, K, L and M are parallel to one another.

Road A is 1 km East of road B. **(Direction and Distance) NIMCET-2014**

Road B is $\frac{1}{2}$ km West of road C.
Road D is 1 km West of road E.
Road G is $\frac{1}{2}$ km South of road H.
Road I is 1 km North of road J.
Road K is $\frac{1}{2}$ km North of road L.
Road K is 1 km South of road M.

6. Which of the following is necessarily true?

- (a) E and B intersect
(b) D is 2 km West Of B
(c) D is atleast 2 km West of A
(d) M is 1.5 km North of L

7. If E is between B and C, then which of the following is false?

- (a) D is 2 km West of A
(b) C is less than 1.5 km from D
(c) Distance from E to B added to distance of E to C is $\frac{1}{2}$ km
(d) E is less than 1 km from A

8. If road E is between B and C, then the distance between A and D is

- (a) less than 1 km
(b) C is 1 km West of D
(c) between $\frac{1}{2}$ km and 2 km
(d) more than 2 km

9. Which of the following possibilities would make two roads coincide?

- (a) L is $\frac{1}{2}$ km North of I (b) C is 1 km West of D
(c) I is $\frac{1}{2}$ km North of K (d) E and B are $\frac{1}{2}$ km apart

10. If K is parallel to I, K is $\frac{1}{2}$ km South of J and 1 km North of G, then which of the following two roads would be $\frac{1}{2}$ km apart?

- (a) I and K (b) J and G
(c) I and G (d) J and K

11. Gopal starts from his house towards West. After walking a distance 30m, he turned towards right and walked 20m. He turned left and after moving a distance of 10m, turned to his left again and walked 40m. Then, he turned left and walked 5m. Finally, he turns to his left. In which direction is he walking now?

(Direction and Distance) NIMCET-2015

- (a) North (b) South
(c) East (d) South East

12. Raghav left his home for office in car. He drove 15 km straight towards North and then turned Eastwards and covered 8 km. He then turned to left and covered 1 km. He again turned left and drove for 20 km and reached office. How far and in which direction is his office from the home? **(Direction and Distance) NIMCET-2017**

- (a) 20 km North -west (b) 15 km North -west
(c) 30 km North -west (d) 25 km North

13. Rishabh stops after going 10 km towards west from his office. Then he goes 8 km turning to his left. After this he goes 4 km turning to his left. How far is he from the fixed point? **(Direction Distance) NIMCET-2020**

- (a) 18 km (b) 8 km (c) 10 km (d) None

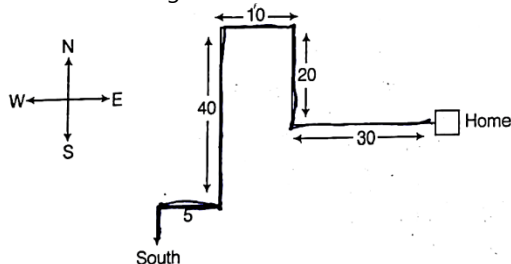
14. On Monday, Akash ran 4 km less than the distance he ran on Tuesday. Sanjay, who ran the same distance on Monday and Tuesday, ran 5 km more on Tuesday than the distance Akash ran on Monday. Find the difference



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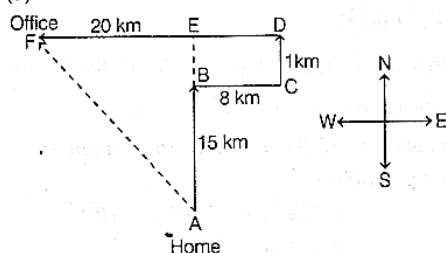


10. (d) J and K would be 1/2 km apart.
11. (b) According to the information given in the question, we can draw a direction diagram.



From the above diagram, it is clear that Gopal is walking in South direction.

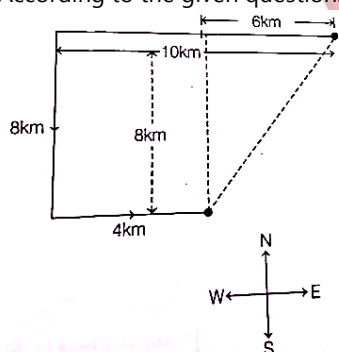
12. (a)



$$\begin{aligned} AF &= \sqrt{(FE)^2 + (AE)^2} \\ \Rightarrow AF &= \sqrt{(20 - 8)^2 + (15 + 1)^2} \\ &= \sqrt{144 + 256} \\ &= \sqrt{400} \\ &= 20 \text{ km} \end{aligned}$$

Raghav's office is 20 km away from his home and in North-West direction.

13. (c) According to the given questions.



Distance between his starting point and final point

$$\begin{aligned} \text{Apply Pythagoras theorem} &= \sqrt{8^2 + 6^2} \text{ km} \\ &= \sqrt{64 + 36} \\ &= \sqrt{100} = 10 \text{ km.} \end{aligned}$$

∴ The correct answer is option (c).

14. (b) Let Akash ran x km on Tuesday.

So, Akash ran (x-4) km on Monday.

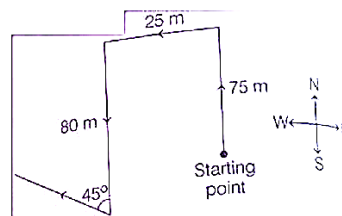
Distance covered by Sanjay on Monday = Distance covered by Sanjay of Tuesday.

$$= x - 4 + 5 = (x+1) \text{ km}$$

Difference between the distances covered by Akash and Sanjay over two days = (2x + 2) - (2x-4)

$$= 2x + 2 - 2x + 4 = 6 \text{ km}$$

15. (c) According to the question,



It can be clearly seen that Deepa is moving in South-West direction.

